ODSC AI

The #1 AI Builders Conference

AGENTIC AI & AUTONOMOUS SYSTEMS

SAN FRANCISCO | OCT 27-29

COGNITIVE MEMORY ARCHITECTURES: DESIGNING CONTINUITY INTO LONG-RUNNING RAG AND AGENTIC SOLUTIONS



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Workshop: https://bit.ly/4yWOKIY

David vonThenen

- Agentic Solutions
- Book In Progress
- AI Harness Wizard
- AI/ML Security Hacker
- Part-time GPU Owner

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Agenda

- Workshop Introduction
  - Setup Environment + Baseline RAG
  - RAG + Real–Time Data + Memory
  - Agentic RAG + RT MCP + Memory
  - Agentic + Memory Lifecycle + Client
- Open Discussion

Section 1:
Workshop Introduction

Breakdown of RAG

Training Data Has Cutoff Date

Retrieval

- Provide Data Beyond Cutoff
- Data Retrieval via "A Method"

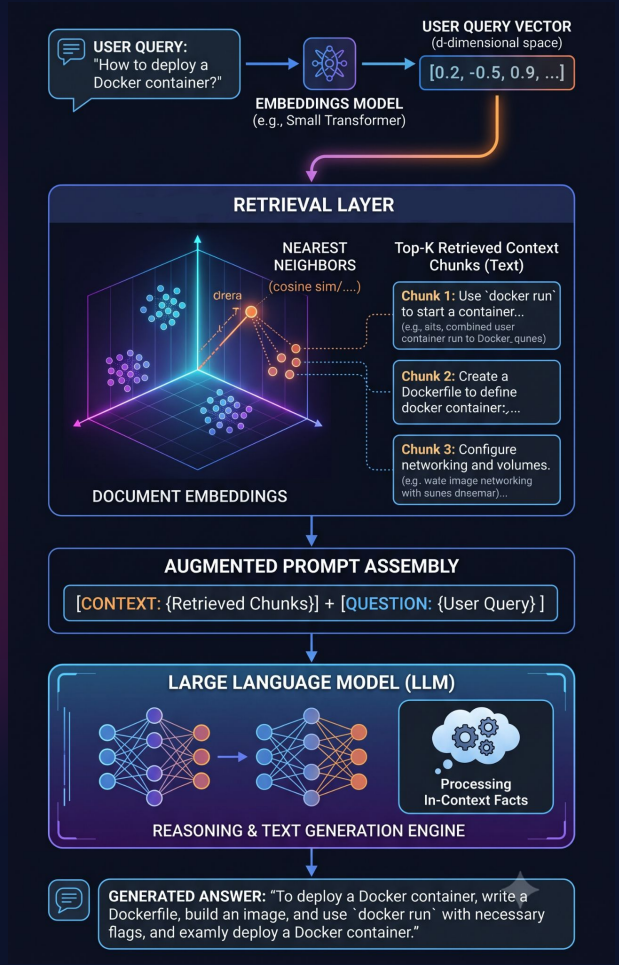
Augmented

- Prompt Assembly
- Provides Context
- Retrieved Data is Truth / Fact

Generation Phase

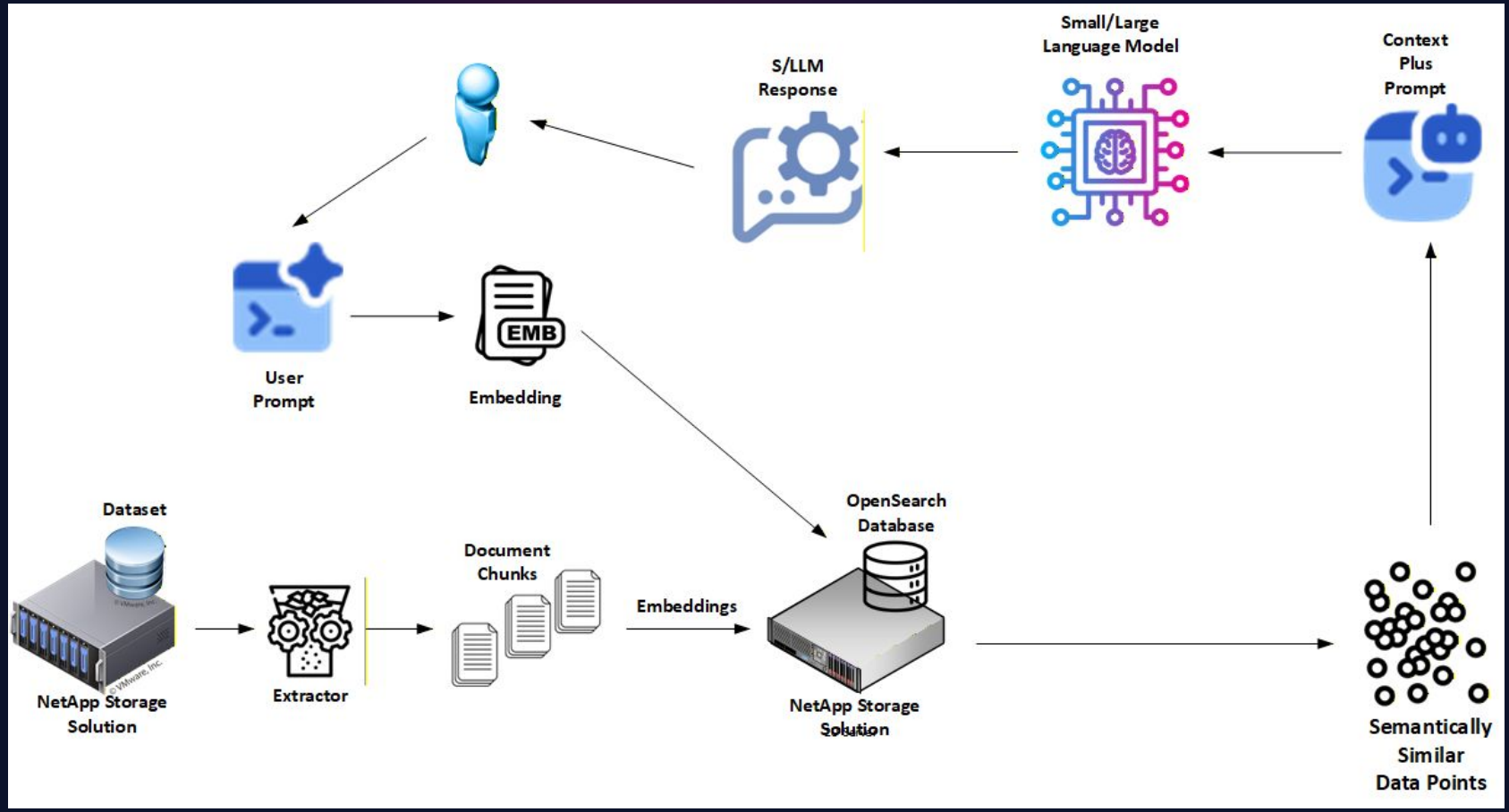
- Lang. Model Synthesizes an Answer

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Gemini Generated



Vector-based RAG

Image Attribution
A David Original



What is Semantic Similarity, Really?

Convert Text to Objects/Vectors:

- Similar Numerical Representation
- Captures Relationships Between:
  - Words, Concepts, etc

Is The Foundation of Vector Search

- Query Nearest Vectors (kNN), like:
  - Cosine Similarity
  - Euclidean Distance

How Does This Relate To Answers?

  - Does Similarity Mean Answers?

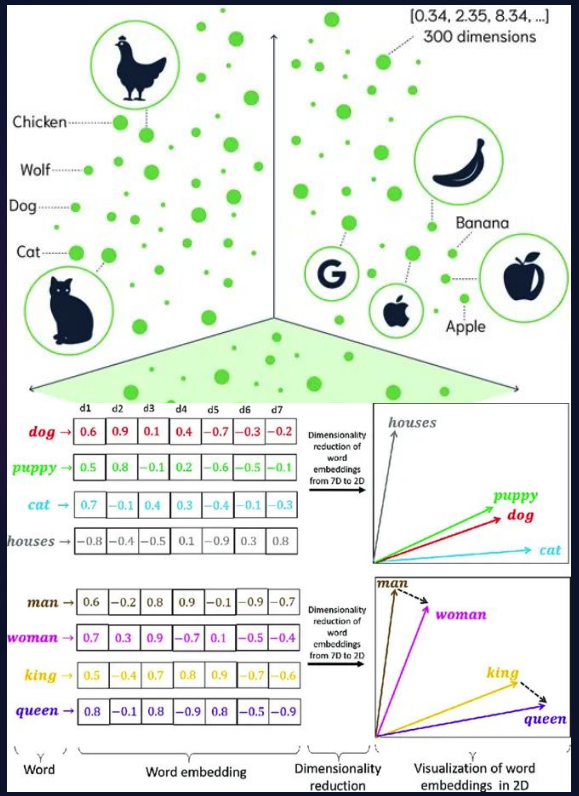
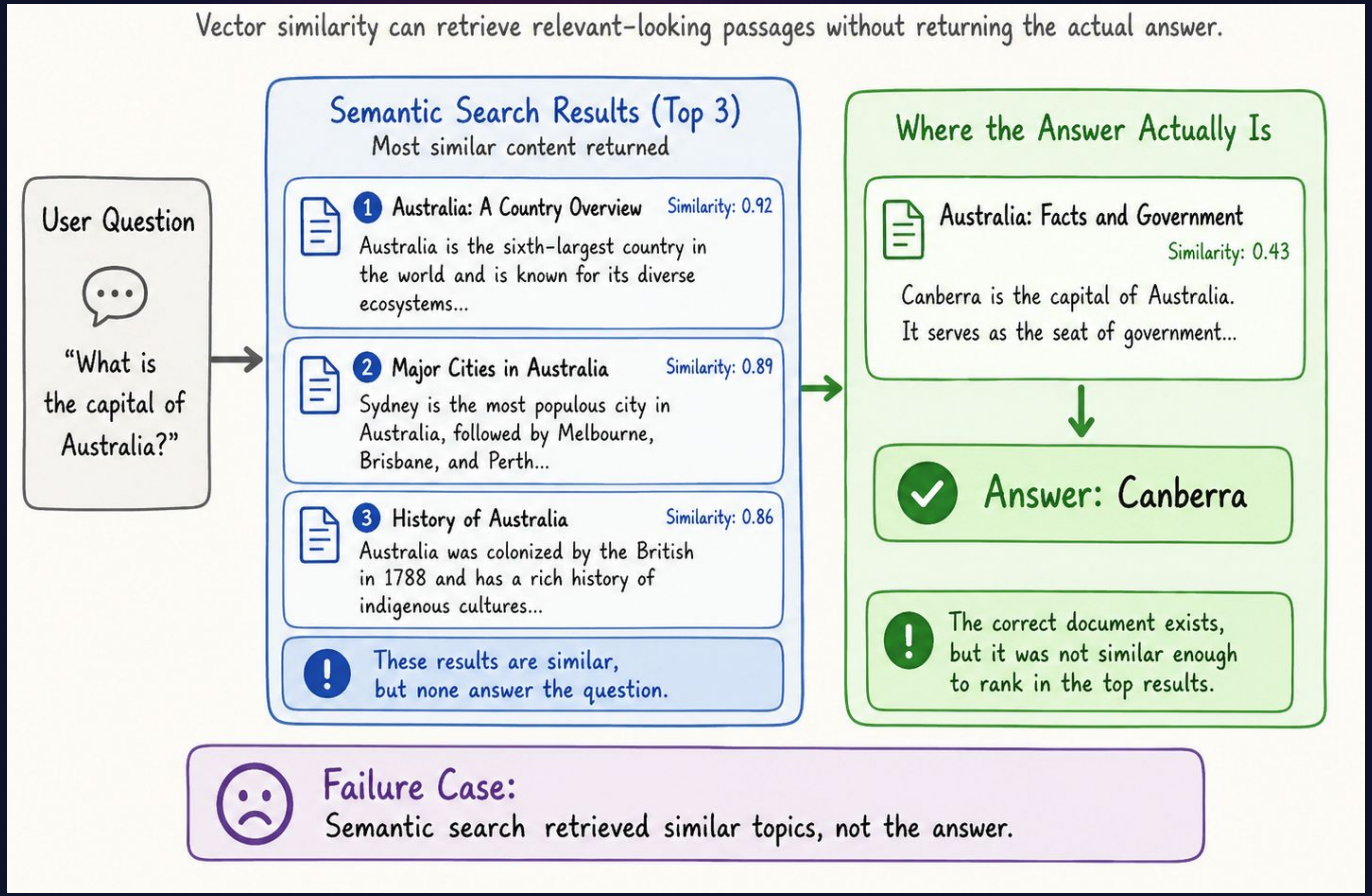


Image Attribution
https://mehmetozkaya.medium.com/understanding-semantic-meaning-and-similarity-search-cosine-similarity-and-euclidean-distance-666fcb5911ea

Similar Context ≠ Answers

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ChatGPT Generated



Hands-On Time!

bit.ly/2026-odsc-west-memory-workshop

Exercise 0: Environment Setup

Exercise 1: Simple RAG Baseline

Observations?



Image Attribution
https://www.astroshop.eu/magazine/practical-tips/tips-and-tricks/5-ways-to-document-your-observations/i,1613

Section 2:
RAG + RT Data + Memory

Agent Memory

Why Agent Memory?

- Tracks User Interaction to the Agent
- Workshop Uses mem0

Memory Enables:

- Continuity: Restart Where You Left Off
- Personalization: User Nuances, Pref., etc
- Episodic Recall: Reuse Past Evidence
- Reusable Context: Use Earlier Findings

What Memory Isn't?

- They Are Not Facts!





Graphiti







ANTHROP\C

What is mem0?

https://github.com/mem0ai/mem0

- Multiple Backends (OpenSearch, pgvector, etc)
- Add: Save Based on Embedding Model
- Search: Semantic Similarity, Supports Tags
- Scope: Multiple Streams (User, Convo, etc)
- Maintenance: Update/Delete Records



Platforms	Claude/OpenAI SDK, Strands, ADK, etc	Easy to use, Tied to specific ecosystems
Graphiti	Knowledge Graphs - NER + Relationships	Relationship, historical change, point-in-time
LangMem	Opinionated Memory Management	Already using LangGraph/LangChain
memU	Wiki-style, shared pref, skill memory	Reusable knowledge across agents/workflows
OpenSearch	Sessions, working, LT, history	Enterprise, auditing, compliance, scale, etc

Real-Time Data Retrieval

Memory Is Recalled, RAG Is Static Info

- Needs Introduction of "Fresh" Data
- Using Tavily (alt: Perplexity Sonar, Querit, etc)

Important Distinction!!

- AI Search Different From Facts
- Using as a Means Of Dynamic Data
- Like Reinforcement Learning, But NOT

Reminder, Reinforcement Learning

- RL: Data is Verified → Dataset
- Workshop: Unvetted Temporary RL Data







Memory for Exercise 2

Complete Convo: `save_complete_conversation()`

- Role-labeled, linked chunks
- Checkpoint of current full snapshot
- Reconstructs conversation continuity

Internet Factoids: `save_internet_results()`

- Web Results With Metadata
- Preserves Evidence Found

Synthesized Conversation: `save_synthesized_turn()`

- User/Agent Turn Saves
- Extraction Candidates of Reusable Knowledge

User/Convo Scope: `MEMORY_USER_ID=user-123, MEMORY_SESSION_ID=482731`

Image Attribution
https://labelyourdata.com/articles/ocr-with-deep-learning

Exercise 2: Architecture

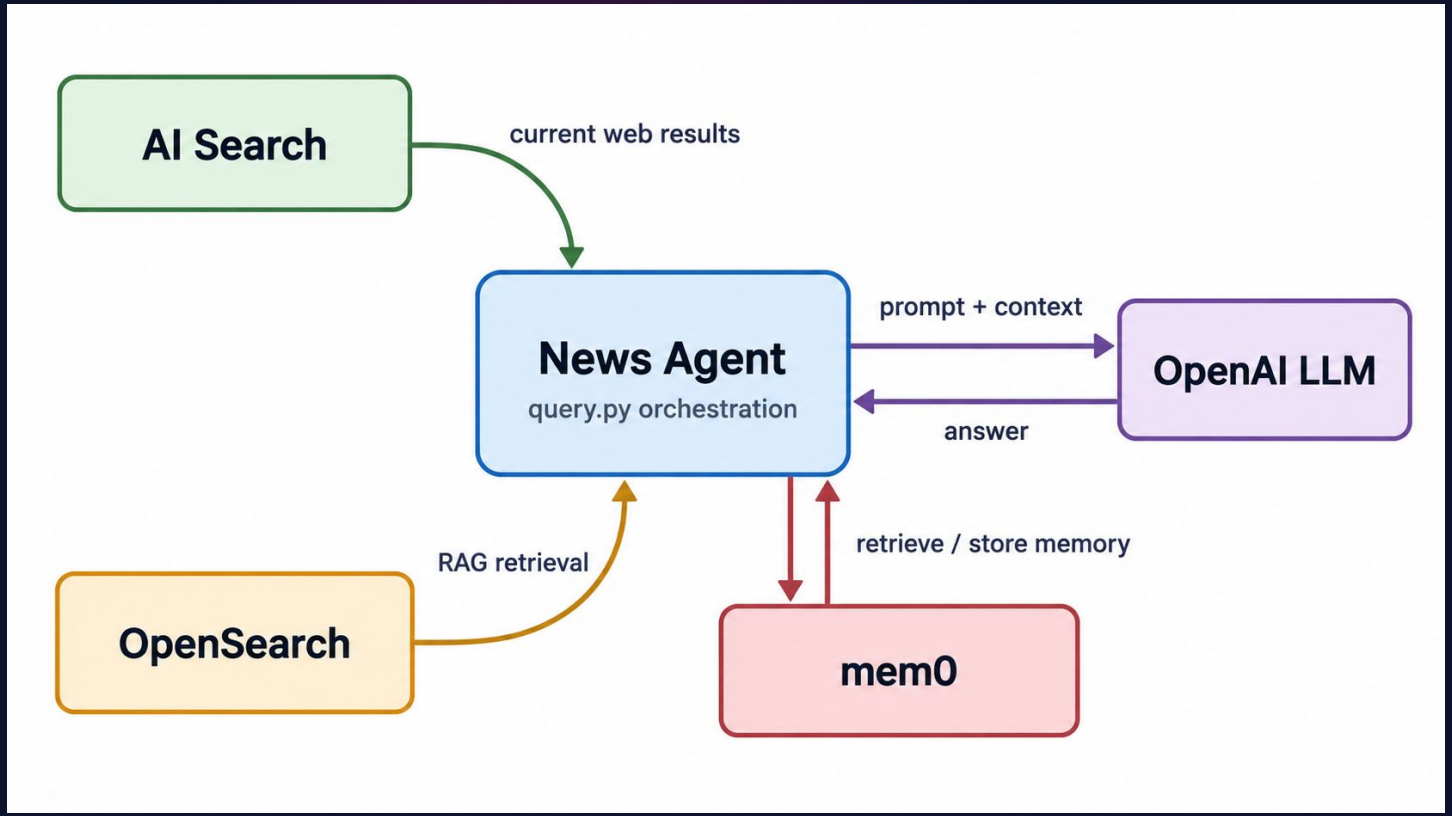


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Memory Recalled

Before Answering:

- Searches Synthesized and Web/Search Memory
- Ranks Semantic Matches and Prefers Named Entities
- Return Relevant/Ranked Memories



Citations for Auditing:

- [DOC #] – Retrieved OpenSearch document chunk
- [WEB #] – Web Result Retrieved for THIS Question
- [MEM #] – Previously Saved, Now Recalled Memory or Evidence
- [TURN #] – Earlier Role-labeled Conversation Message

Provenance!! We Know Where a Claim Came From!!

Image Attribution
https://machinelearningmastery.com/ai-agent-memory-explained-in-3-levels-of-difficulty/

Hands-On Time!

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Exercise 2 Combines:

- Embedding Retrieval (RAG)
- Real-Time Data
- Simple Agent Memory

Observations?



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https://teachingentrepreneurship.org/teaching-customer-observation/

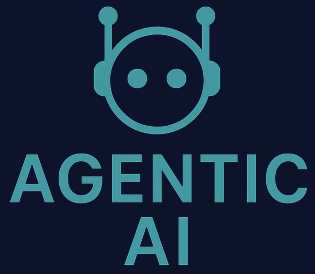
Section 3:
Agentic RAG + RT Data + Mem

Level Up: Agentics, MCP, & More

Let's Iterate! Improvements in this Exercise...

- Model Assisted Decision With:
  - Course of Action & Memory Selection!
- AI Search Relocated to MCP
- Memory Integration Includes:
  - Synthesis of Data
  - Verification of Input/Response
  - Real Governance
    - Data, Turns, etc







What Make This Agentic?

Sequential Flow → Non-Deterministic w/ Policy

- Policy Can Include Multiple MCP Calls
- Context → Synthesis → Resp w/ Policy
  - Context = Historical + Embed/Fresh
- Memory The Same, Exception:
  - Factual History via Embedding
  - Real-Time vs MCP/Tavily
- Big Difference: Policy Enforcement
  - Permitted Retrieval Paths
  - Synthesis with Governance

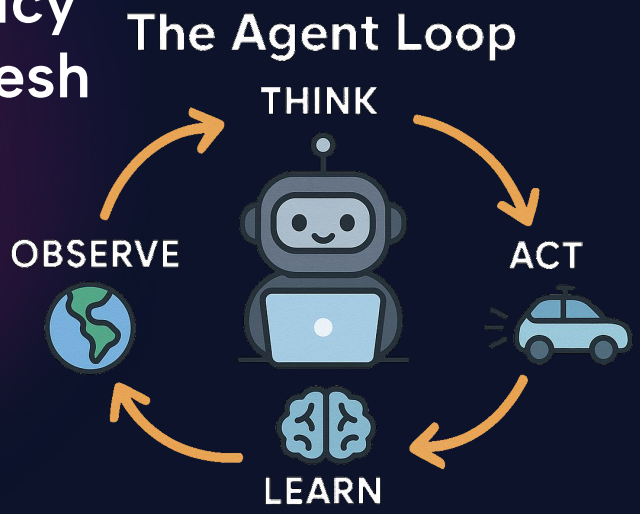


Image Attribution
https://medium.com/towards-explainable-ai/inside-the-agent-loop-the-core-of-autonomous-ai-systems-ab6f79e7baf0

Synthesis of Data and Response

Two Non-Deterministic Paths

- Orchestration using gpt-5.4
- Synthesis/Data/Verify using gpt-5.4

Ideally, Using Two Different Models

- Clear Separation of Roles
- Orchestrator Only Affects Workflow
- Synthesis Only Affects Responses



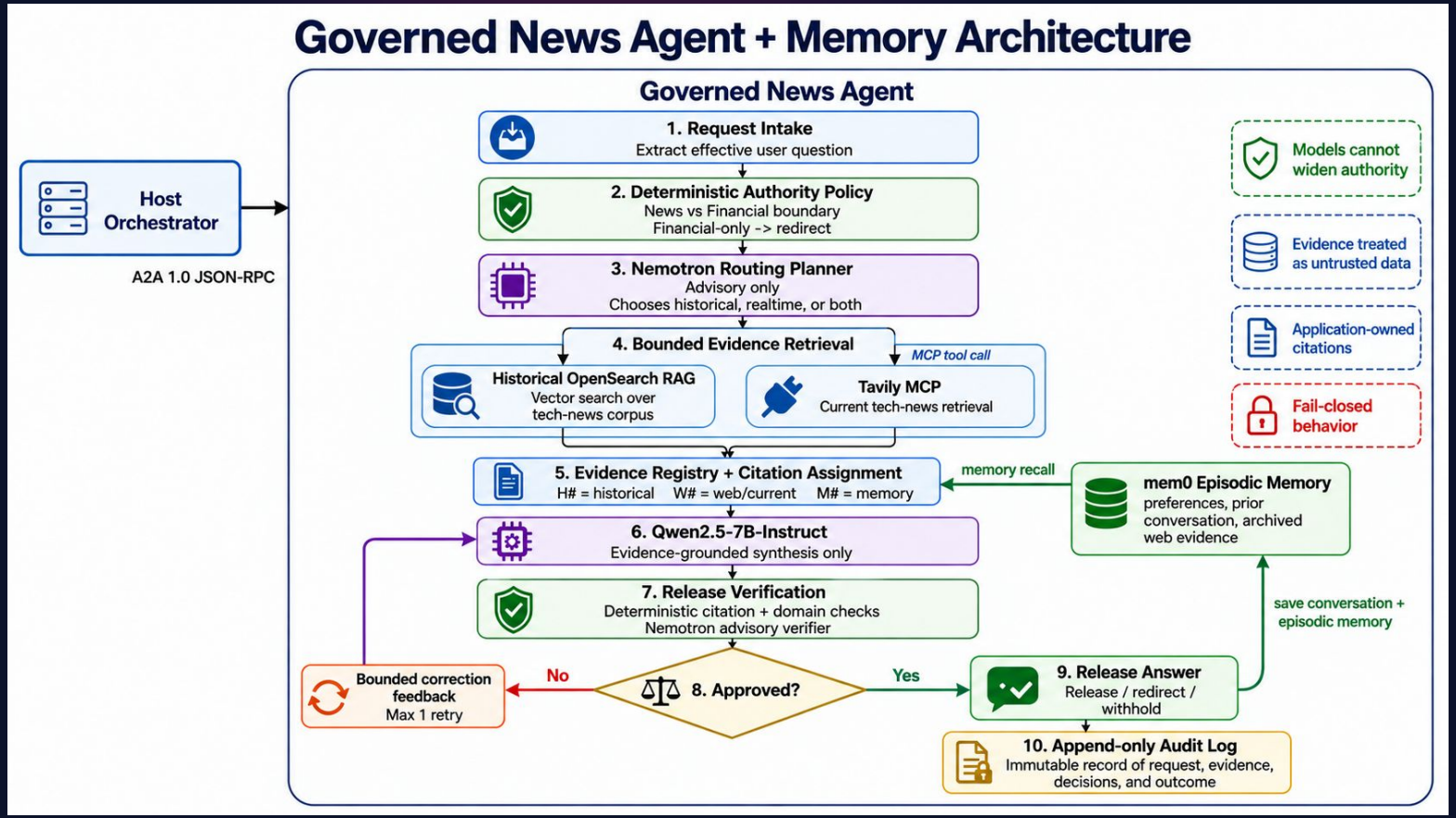
Policy States:

- Verify Claim, Citation IDs, No Financial Claims

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Gemini Generated

Workflow: Agentic News Expert

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ChatGPT
Generated



Follow The Memory

Memories Saved in Agent:

- Complete Conversation
- Synthesized Memory
- Internet "Episodes" (Not Facts)

Using mem0 Python SDK:

- All Memory:
  - Labeled Using NER
  - mem0 SDK: `infer=False`
- Conversation/Turns:
  - Labeled Using NER
  - mem0 SDK: `infer=True`



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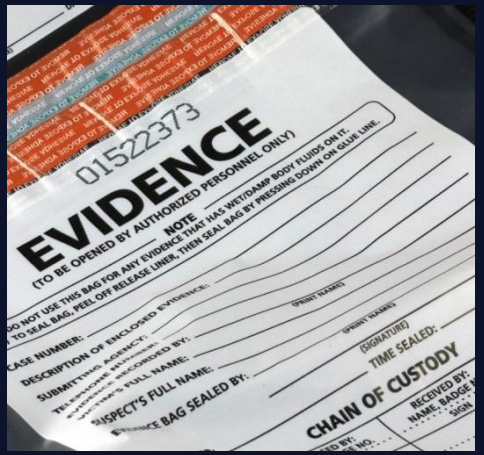
Memory and Response Synthesis

Before Answering:

- Retrieve Historical (RAG using OpenSearch)
- Obtain Real-Time Data (MCP using Tavily)
- Recall RELEVANT Conversational Memory (mem0)

Citations for Auditing:

- [H#] – Retrieved OpenSearch document chunk
- [W#] – Web Result Retrieved for THIS Question
- [M#] – Previously Saved, Recalled Memory or Evidence



Provenance!! Understand Origins of the Claim!!

THEN SYNTHESIZE A RESPONSE w/ POLICY!!

Image Attribution
https://labelyourdata.com/articles/ocr-with-deep-learning

Hands-On Time!

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Exercise 3 Combines:

- Agentic Loop
- Embedding Retrieval (RAG)
- Real-Time Data (MCP)
- ”Simple” Agent Memory

Observations?



Image Attribution
https://www.spaceclick.com/blog/introduction-to-astronomical-observation-instruments-and-deep-space-photography/

Section 4:
Agentics w/ Memory Lifecycle

Let's Talk Memory Lifecycle

Agentic RAG + RT Data with Full Memory Lifecycle

- Recall Raw Convo + Topic → Versioned History
- No Working Summary → Memory Compaction
- Memory is Forever → Memory Expiration
- Request, Memory Audit → Lifecycle Events, Summary

Only Adding Agentic Loops!

What's Changed:

- How Memory is Saved
- Memory Can Be Transformed
- Update and "Forget" Memory

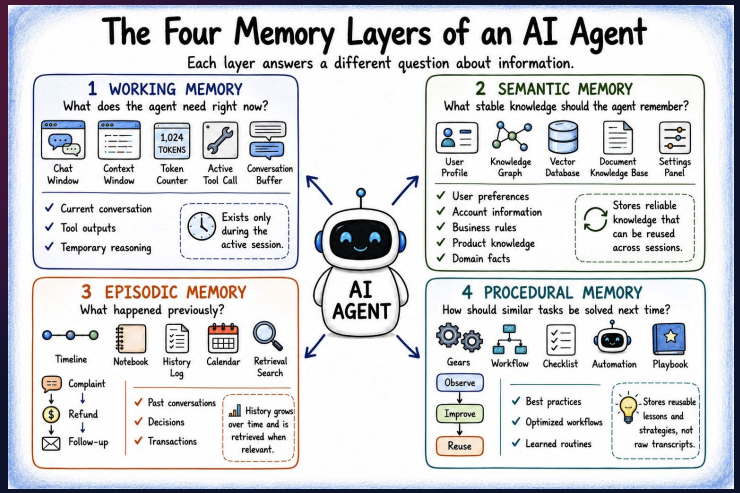


Image Attribution
https://machinelearningmastery.com/choosing-the-right-ai-agent-memory-strategy-a-decision-tree-approach/

Automatic Convo Compaction

Client Implements `messages[]` on OpenAI API

- To Accelerate Compaction, `compact_k = 5`

Q&A Turn-By-Turn Flow:

- Client Sends Message/Prompt
- If `len(messages[]) > compact_k` Then:
  - Generate One Continuity Summary
- Publish New Convo Version

High Turnover is Intentional on Demo!

- Real World: Tune `compact_k` to Context Window

Conversation Compaction

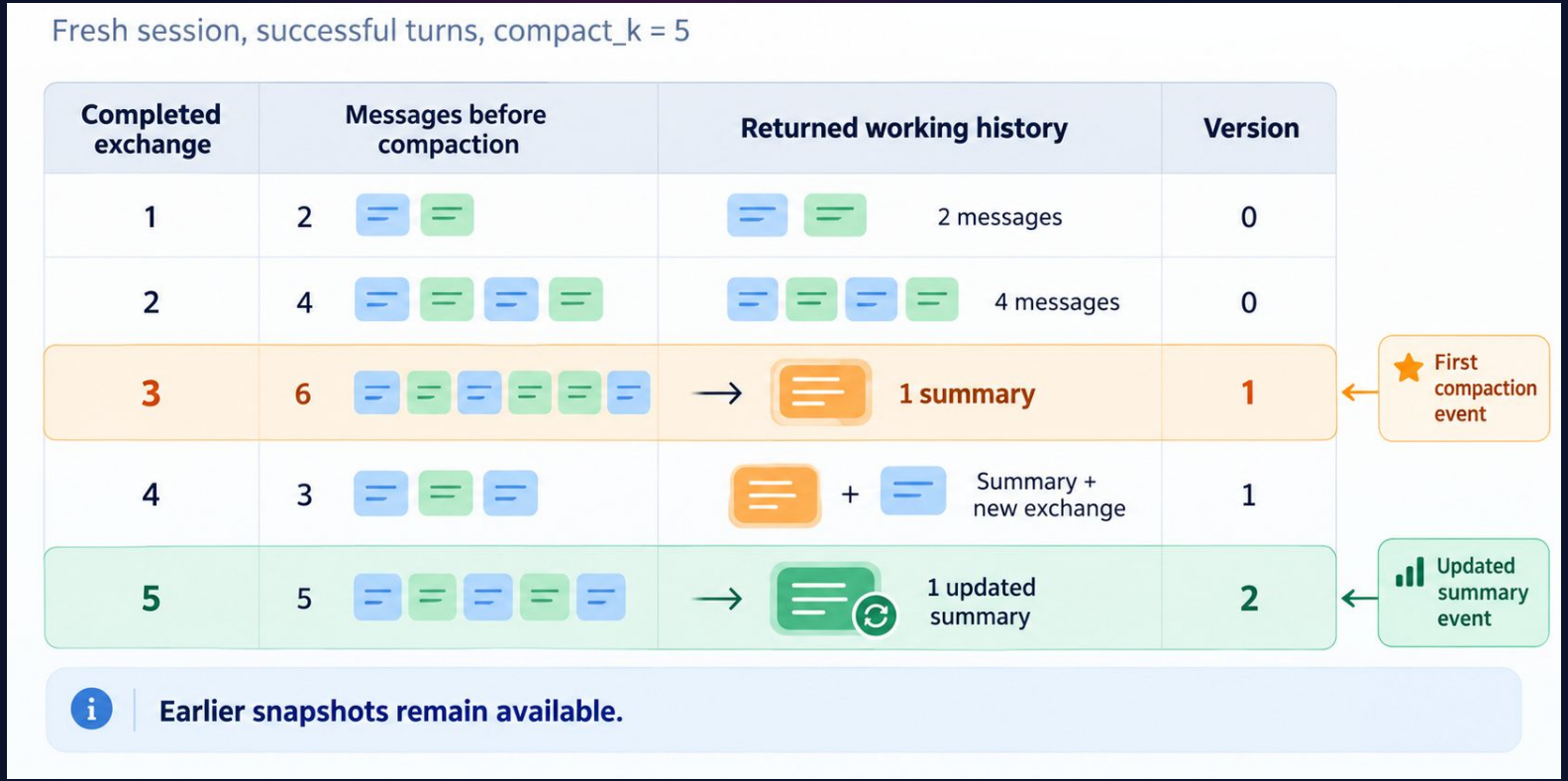


Image Attribution
ChatGPT

Memory Expiration

What Receives a UTC Expiration?

Inferred Q&A Memory	Excluded From Topic Recall
AI Search Data (Tavily)	Excluded From Topic Recall
Raw Conversation	Retained / Selectively Included
Working History & Snapshots	Retained / Selectively Included

Memory Summary:

- New Inferred (Q&A and RT Data) Receive UTC Expire
- Reads Leave That Deadline Unchanged
- Expired Topic Records Remain Stored

Big 5 Memory Operations

1. Capture: Save Approved Turns, Snapshots, & Data
2. Retrieve: Restore History & Sligible Memories
3. Update: Advance History via Data Updates
4. Consolidate: Publish Compacted Working Version
5. Forget: Exclude Expired Topic-Memory Candidates

Remember, With AI Governance!

- Tracing, Auditing, Observability, etc
- When Something Goes Wrong, How to Debug?

Other Use Cases

Potential Other Enhancements:

- Memory Decay (Less Effective Over Time):
  - 0 Day = 100%, 7 Days = 50%, 14 Days = 25%, etc
- Different Memory Policies
  - Set Based on Your Needs
  - Multiple Data Sources
  - Different Policies
- Reinforcement Learning + Memory
  - Data Tiering, Promotion, Demotion, etc



Image Attribution
https://www.psychologistworld.com/memory/peterson-decay

Hands-On Time!

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Exercise 4 Combines:

- Agentic Loop
- Embedding Retrieval (RAG)
- Real-Time Data (MCP)
- Agentic Memory (Full Lifecycle)

Observations?



Image Attribution
https://stockcake.com/i/lab-experiment-observation_84386_9511

Resources

Resources

All Materials (Slides, Code, Instructions, etc):

github.com/davidvonthenen/2026-odsc-west

Things To Check Out:

Free AI Courses

www.ntap.fyi/courses

Launching Instaclustr Labs

github.com/instaclustr.com/labs



And More Resources

All Materials (Slides, Code, Instructions, etc):

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Vertical APIs for Agentic AI: Routing the Right Context to the Right Expert

- YouTube Recording
- GitHub Resources (Demo, etc)

Less Compute More Impact How Model Quantization Fuel the Next Wave of Agentic AI

- YouTube Recording
- GitHub Resources (Demo, etc)



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Thank You!



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